

The Working Cell

Unit 1; Chapter 5



Learning Objectives 1

- Define the terms energy, kinetic energy, potential energy, and the principle of conservation of energy.
- Define entropy and describe the relationship between heat and entropy.
- Explain how ATP powers cellular work and how ATP is recycled.

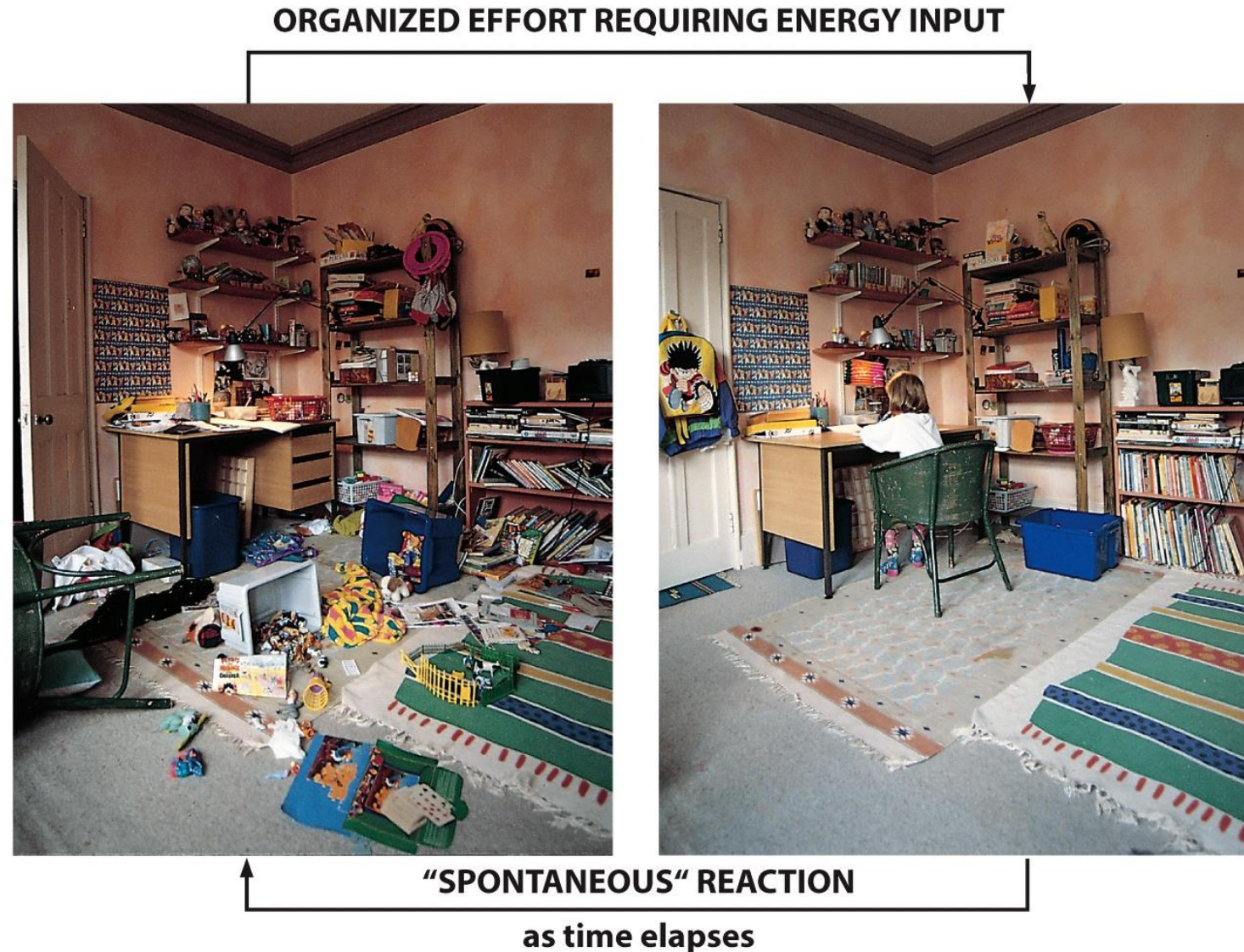
Energy, the Capacity to Cause Change

- Kinetic energy, the energy of motion.
- Potential energy is stored energy, energy that an object has because of its
 - location
 - structure
- Life depends on countless similar conversions of energy from one form to another.

Law of Thermodynamics

- The first law of thermodynamics, known as **conservation** of energy
 - It is not possible to destroy or create energy.
 - Energy can only be **converted** from one form to another.
- Second Law of Thermodynamics
 - the universal tendency of things to become disordered
 - the degree of disorder is always increasing in the universe
 - **Entropy** is a measure of the amount of **disorder**, or **randomness**, in a **system**.

Why human body is highly organized when the disorder of the universe is always increasing?



Heat, the energy in its most **disordered, chaotic** form the energy of aimless molecular movement

- All energy conversions generate some heat, difficult to harness.

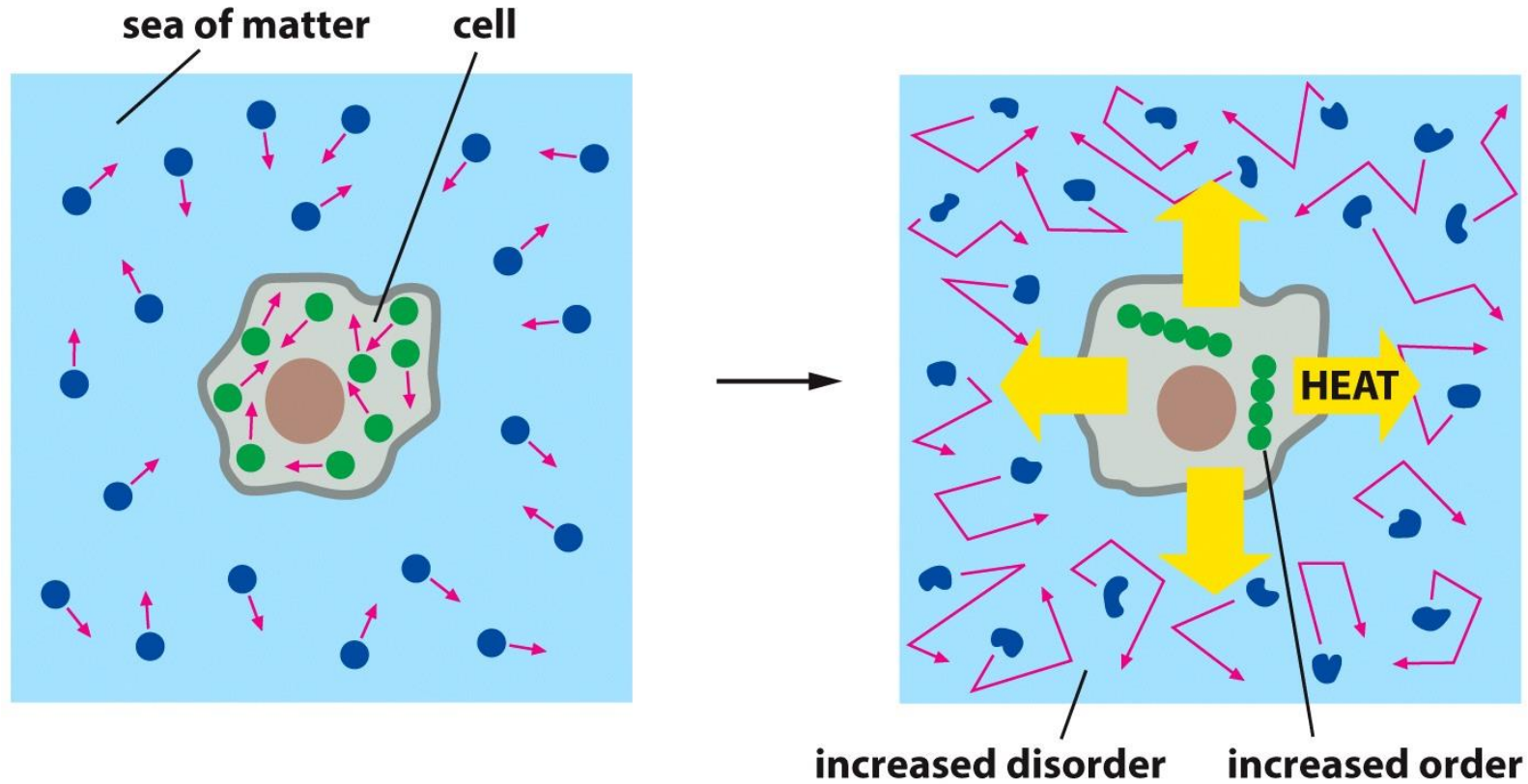
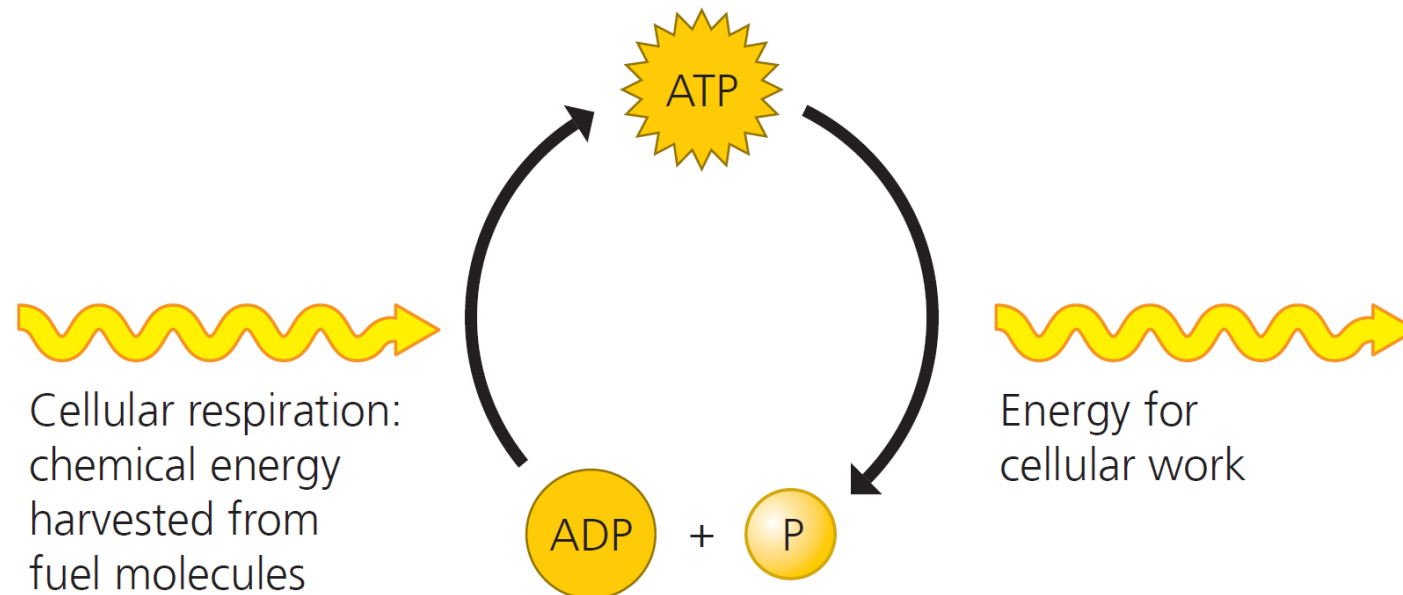


Figure 3-5 Essential Cell Biology, 4th ed. (© Garland Science 2014)

ATP, Adenosine Triphosphate (三磷酸腺苷) for Energy Transformations

- Acts like an energy shuttle
- Stores energy obtained from food
- Releases it later as needed
- Such energy transformations are essential for all life on Earth.



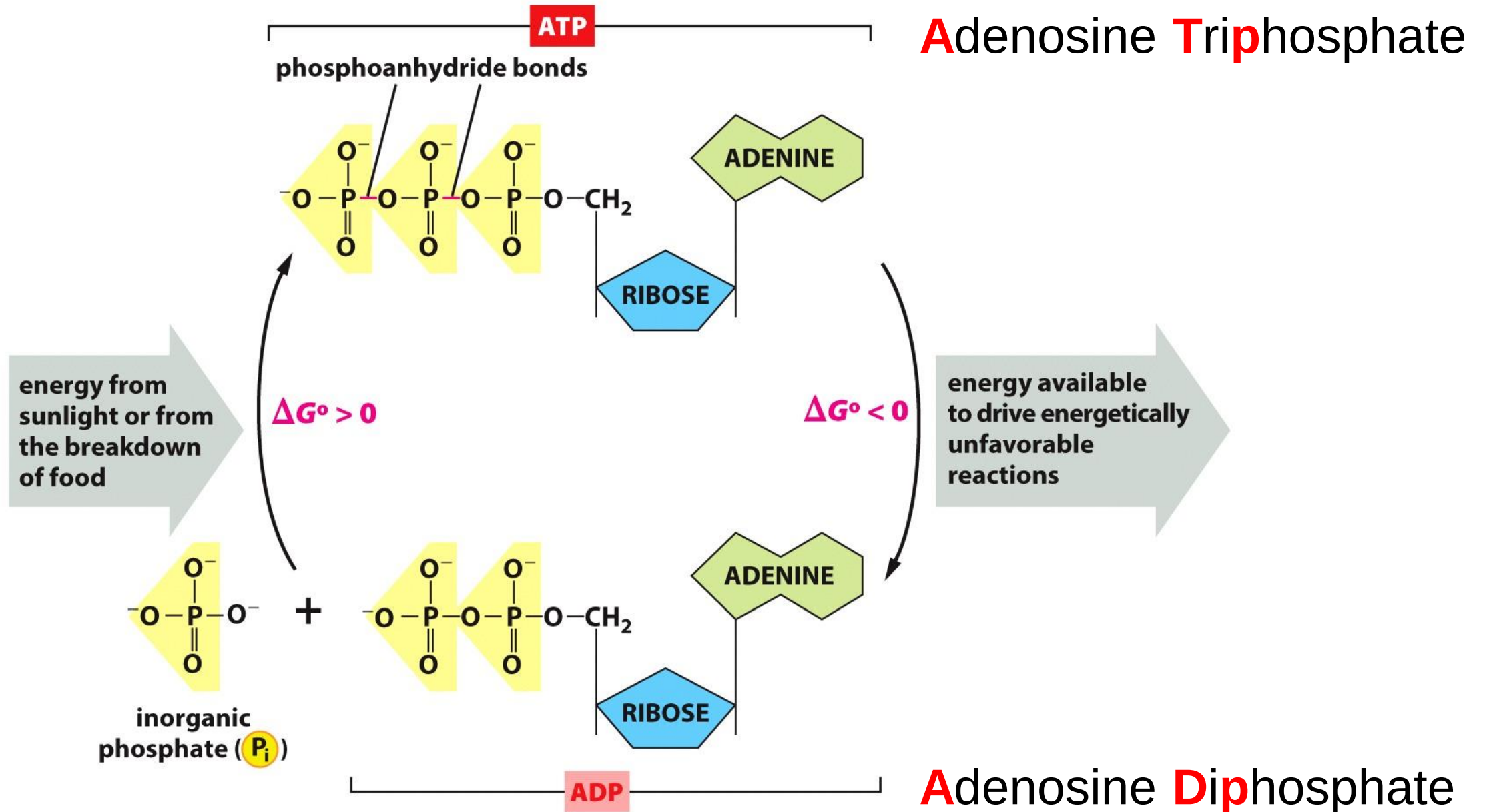


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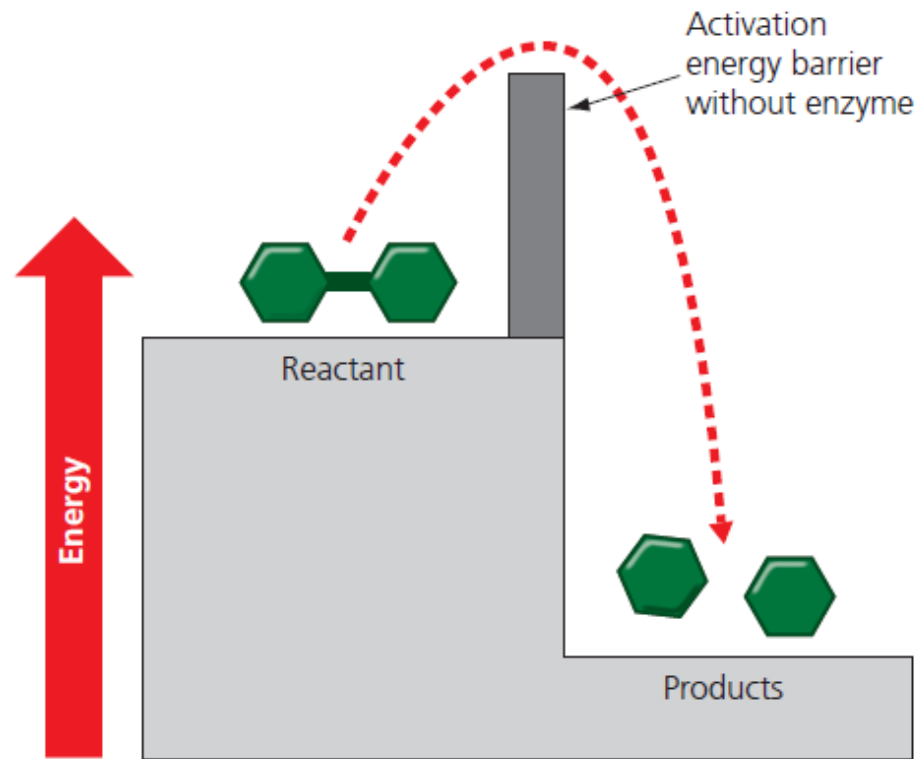
Enzymes

LO 2:

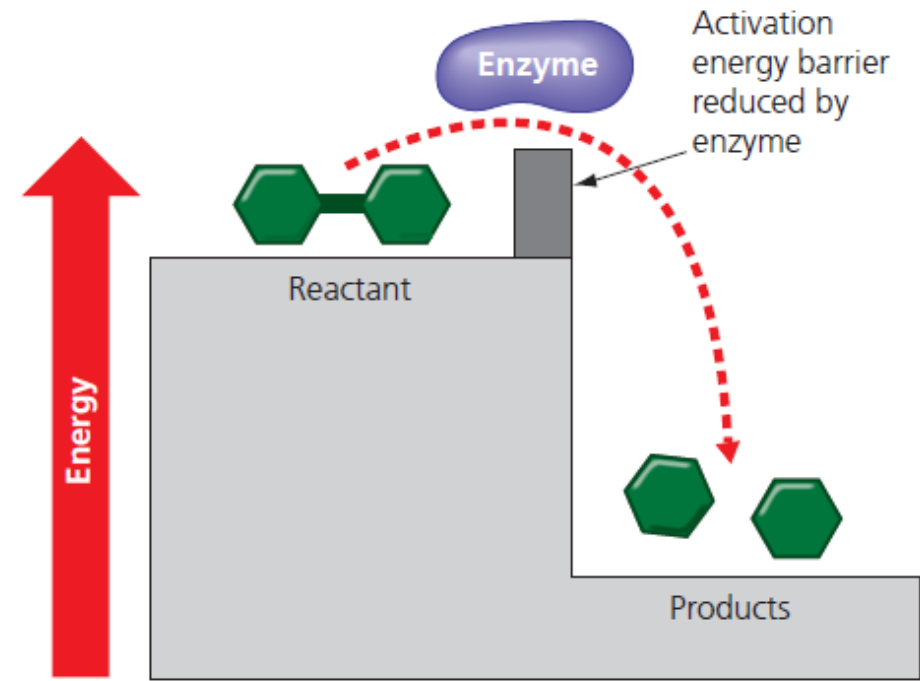
- Explain why enzymes are needed in living organisms and how they speed up specific chemical reactions.
- Explain how inhibitors and poisons can affect enzyme activity.
- Metabolism (新陈代谢) is the total of all chemical reactions in an organism.
- Most metabolic reactions require the **assistance of enzymes**, proteins that **speed up** chemical reactions without being consumed by the reaction.
- All living cells contain thousands of different enzymes, each promoting a different chemical reaction. (**Highly Selective**)
- Many enzymes are named for their substrates, but with an **–ase** ending.

Enzymes Function by Reducing the Activation energy

- Activation energy is the energy that must be invested to start a reaction, by activating the reactants triggering a chemical reaction



(a) Without enzyme. A reactant molecule must overcome the activation energy barrier before a chemical reaction can break the molecule into products.



(b) With enzyme. An enzyme speeds the chemical reaction by lowering the activation energy barrier.

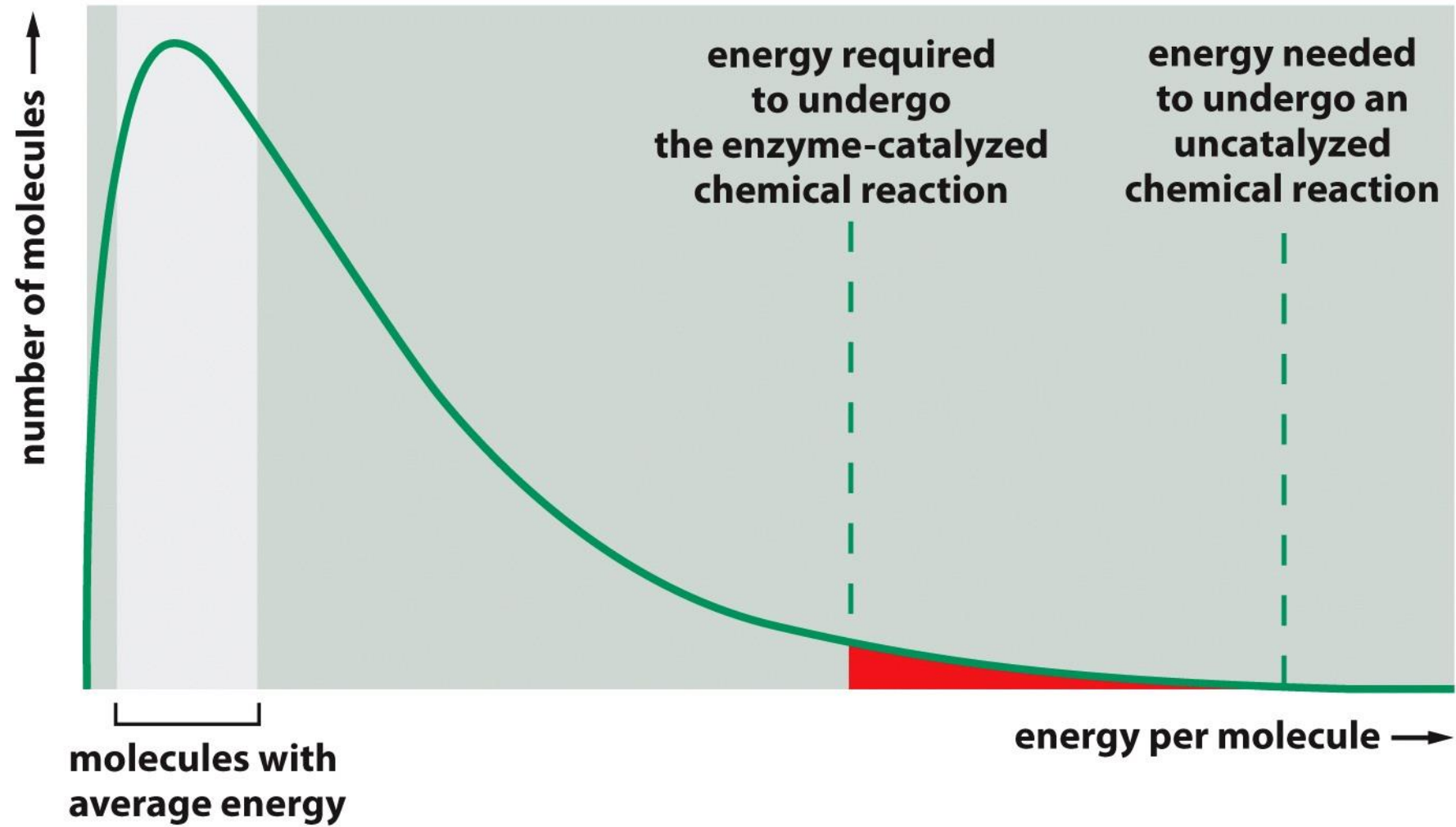
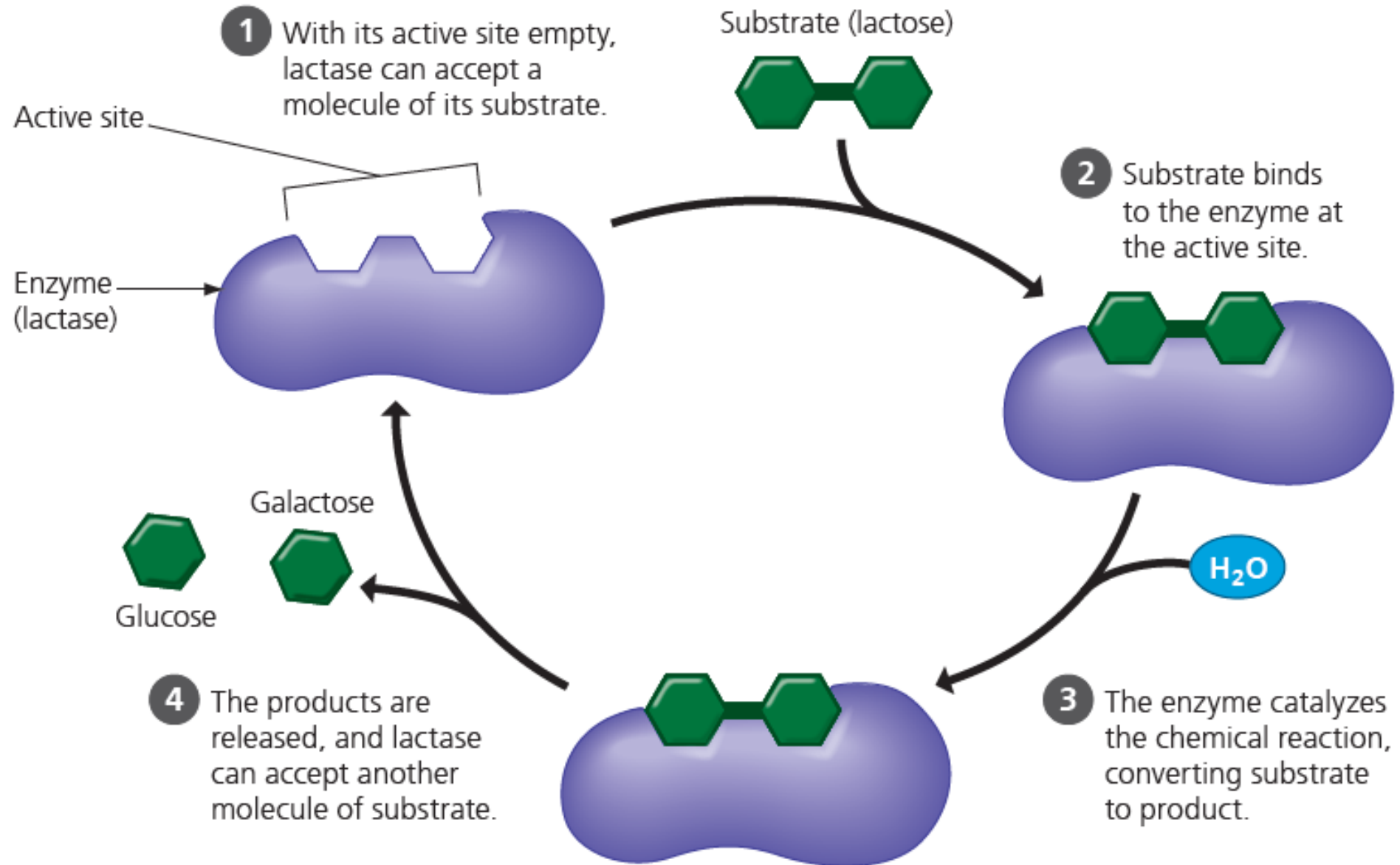


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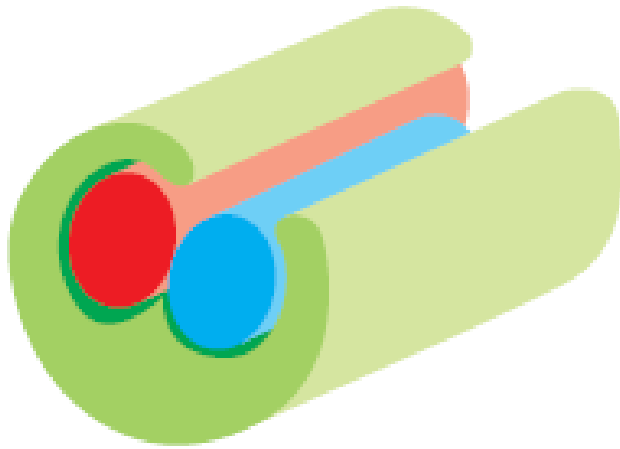
Highly Selective Enzymes

- Each enzyme recognizes a substrate, a certain reactant molecule with the **active site** bearing the **shape and chemistry** that fits the substrate molecule.
- **Induced fit** (诱导配合), the entry of the substrate induces the enzyme to **change shape slightly**, making the fit between the substrate and active site snugger.
- After the products are released from the active site, the enzyme can accept another molecule of its substrate.

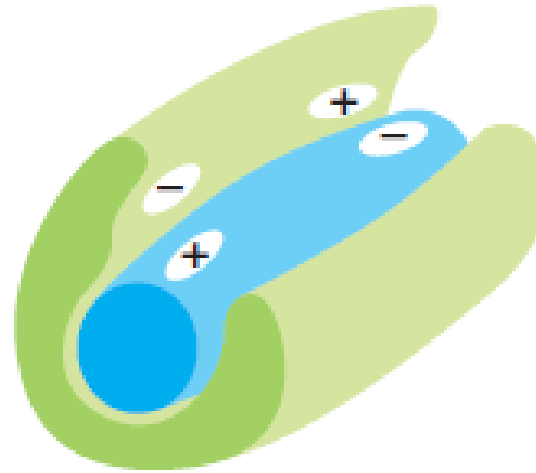
Enzymes Function Repeatedly



Ways for Enzymes to Favor a Reaction



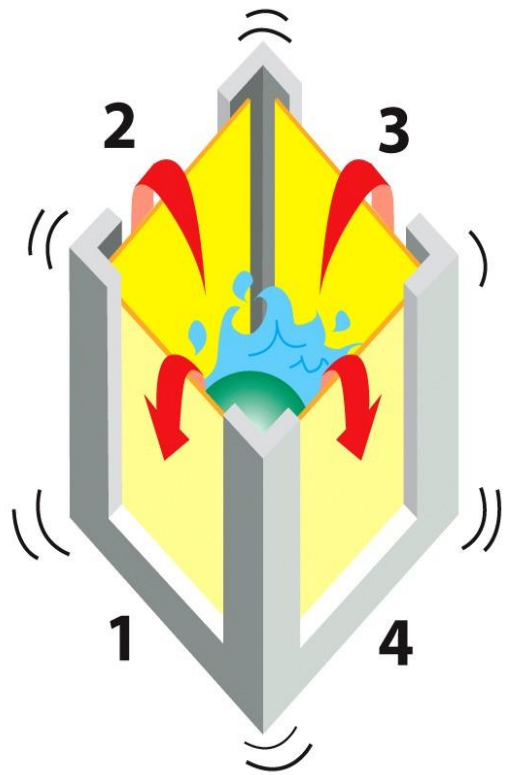
(A) enzyme binds to two substrate molecules and orients them precisely to encourage a reaction to occur between them



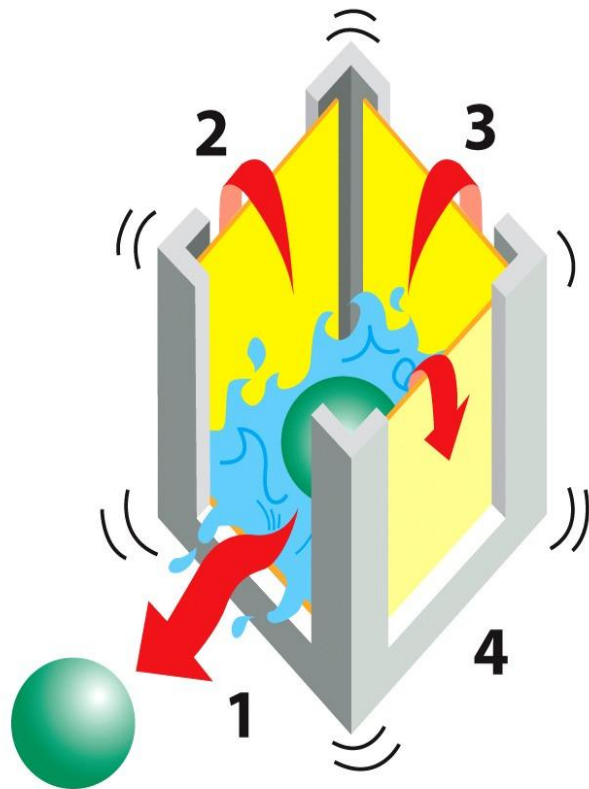
(B) binding of substrate to enzyme rearranges electrons in the substrate, creating partial negative and positive charges that favor a reaction



(C) enzyme strains the bound substrate molecule, forcing it toward a transition state that favors a reaction



uncatalyzed



**enzyme catalysis
of reaction 1**

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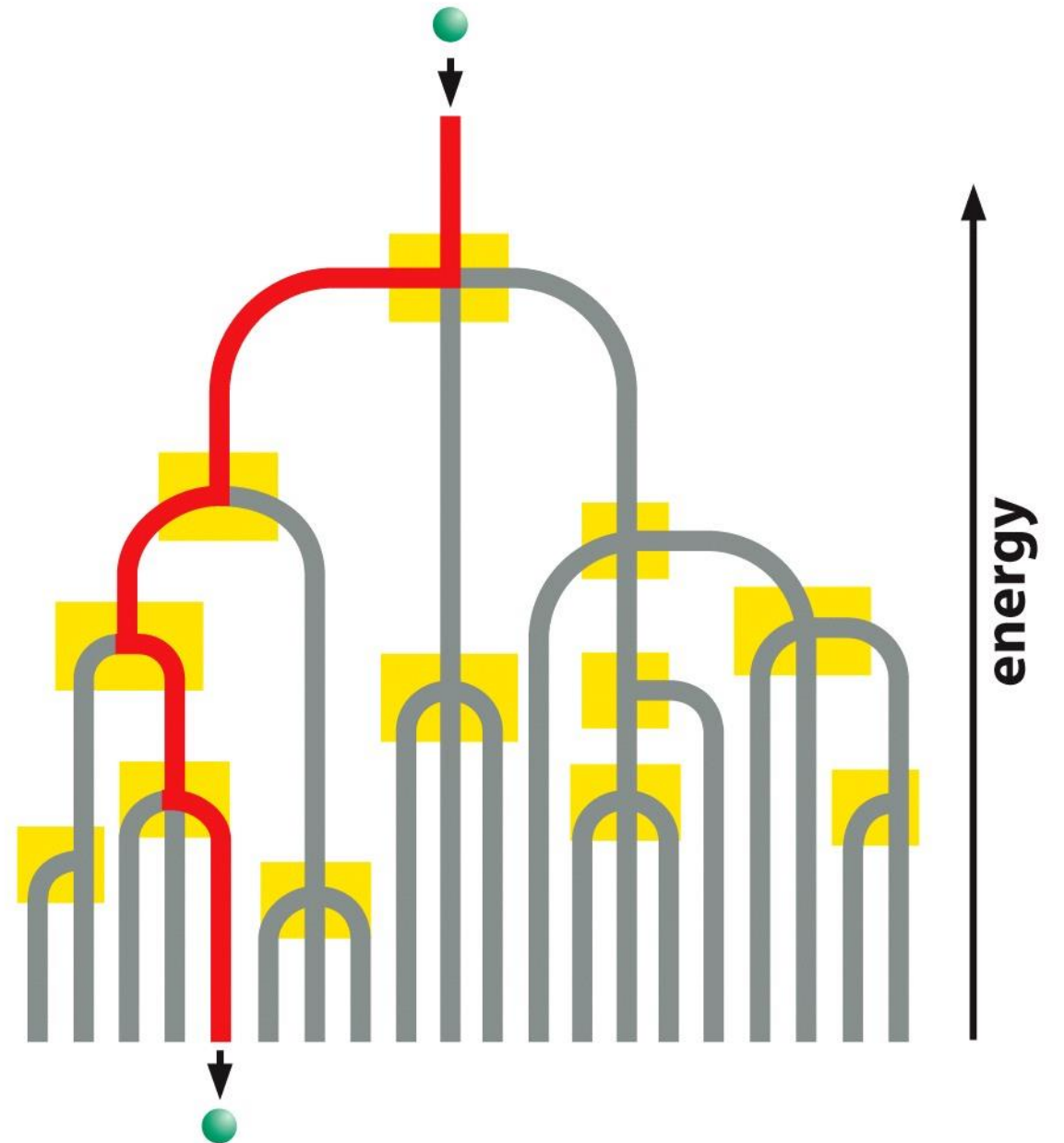
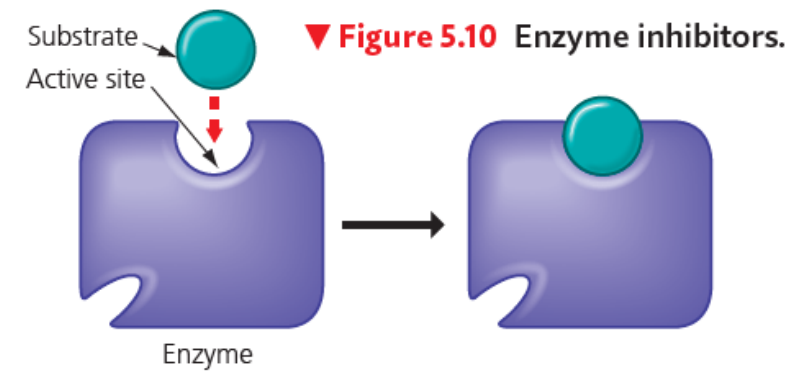


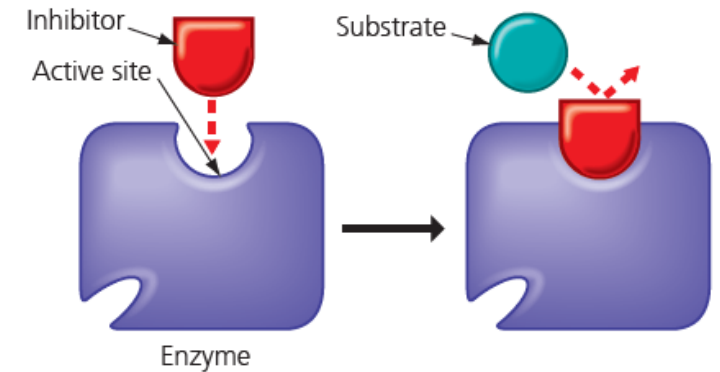
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Enzyme Inhibitors (酶抑制剂)

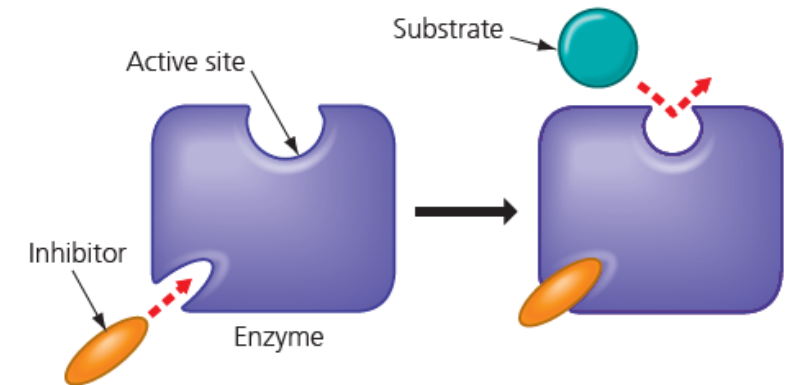
- Competitive inhibitors
 - Substrate imposters, plug up the active site
- Non-competitive Inhibitor
 - bind at a different site from the active site
 - the active site no longer accepts the substrate.
- Activity of almost every cell enzyme is regulated by feedback inhibition.



(a) Enzyme and substrate binding normally



(b) Enzyme inhibition by a substrate imposter



(c) Inhibition of an enzyme by a molecule that causes the active site to change shape

Drugs as Enzyme Inhibitors

- Penicillin (青霉素) blocks the active site of an enzyme that bacteria use in making cell walls.
- Ibuprofen (analgesics) (布洛芬(消炎镇痛药)) inhibits an enzyme involved in sending pain signals.
- Many cancer drugs inhibit enzymes that promote cell division.
- Many toxins and poisons also work as inhibitors.
 - Cyanide inhibits mitochondrial enzyme cytochrome c oxidase, key enzyme in cellular respiration.

Deep Learning Models for Predicting Human Cytochrome P450 Inhibition and Induction

Zhaodi Xiao* and Hajime Hirao*



Cite This: *J. Chem. Inf. Model.* 2025, 65, 9947–9961



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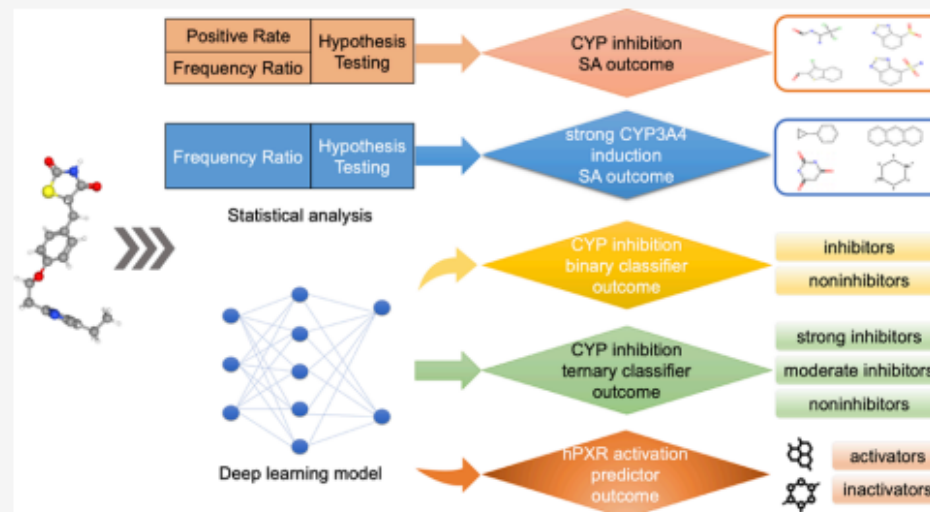


Article Recommendations



Supporting Information

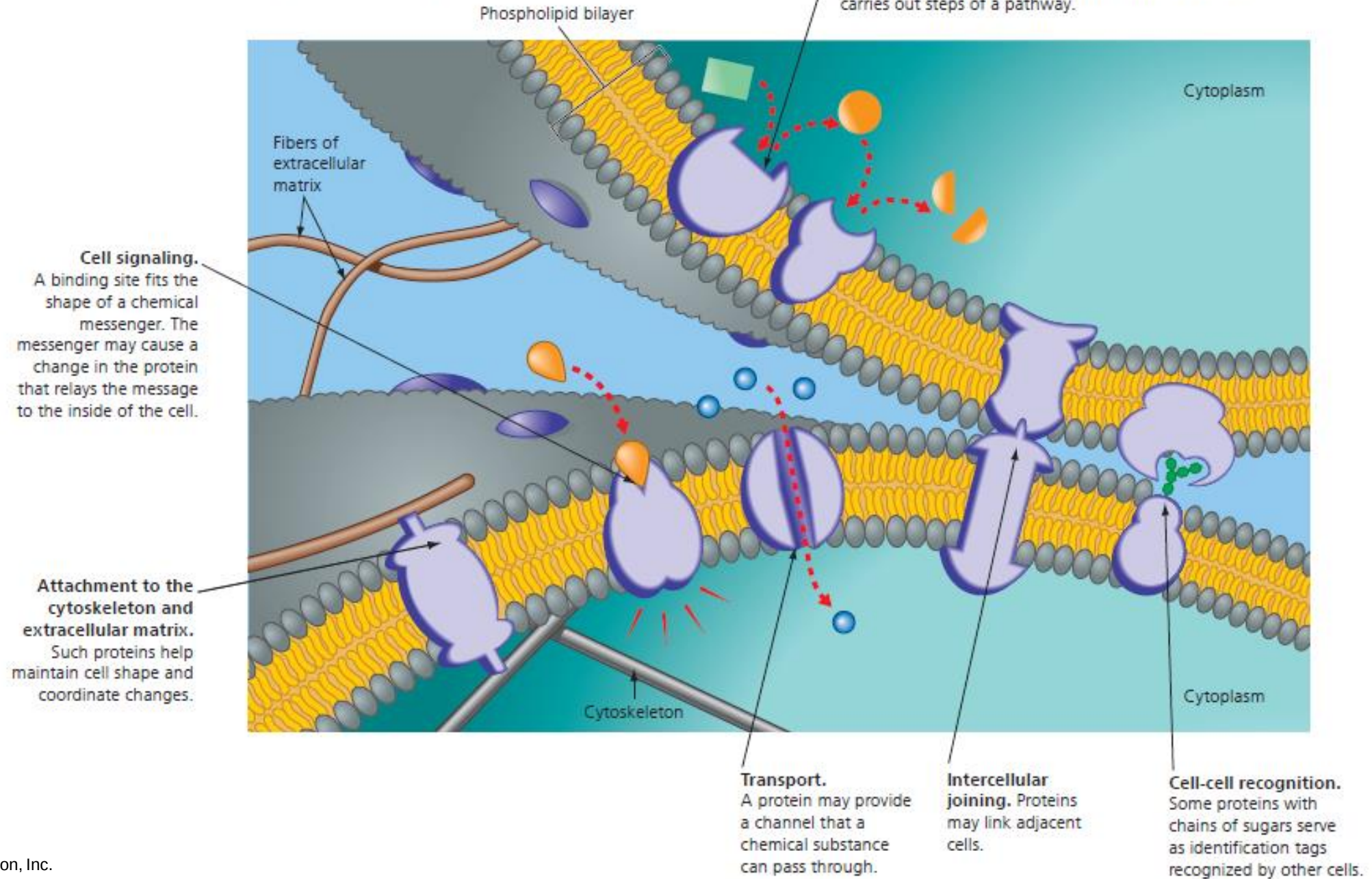
ABSTRACT: Given the critical roles played by human cytochrome P450 enzymes (CYPs) in drug metabolism, accurately predicting their potential inhibition and induction by drugs and drug candidates is a key objective for improving drug development and safety assessment. Traditional experimental methods for identifying CYP modulators are labor-intensive and costly, underscoring the need for efficient in silico prediction models. In this study, we present an advanced deep learning model for predicting CYP inhibition, with a primary focus on key enzymes involved in drug metabolism: CYP3A4, CYP2D6, CYP1A2, CYP2C9, and CYP2C19. This model integrates deep neural networks with principal component analysis (PCA) and the synthetic minority oversampling technique (SMOTE), and it demonstrates excellent predictive performance. Furthermore, we developed a novel classification model capable of accurately distinguishing compounds as strong inhibitors, moderate inhibitors, or noninhibitors for these CYPs, achieving robust and reliable overall performance. Through statistical analysis, we also identified structural alerts (SAs) associated with CYP inhibition and strong CYP3A4 induction, providing a more precise characterization than previous approaches. Finally, we introduced a novel deep learning-based method specifically designed to predict human pregnane X receptor (hPXR) activation, a major mechanism responsible for CYP induction, which also achieved good performance.



Learning Objectives 3

- Define and distinguish between the following pairs of terms:
 - diffusion versus osmosis
 - passive transport versus active transport
 - hypertonic versus hypotonic
 - endocytosis versus exocytosis
- compare water balance between plant and animal cells.

▼ **Figure 5.11 Primary functions of membrane proteins.** An actual cell may have just a few of the types of proteins shown here, and many copies of each particular protein may be present.



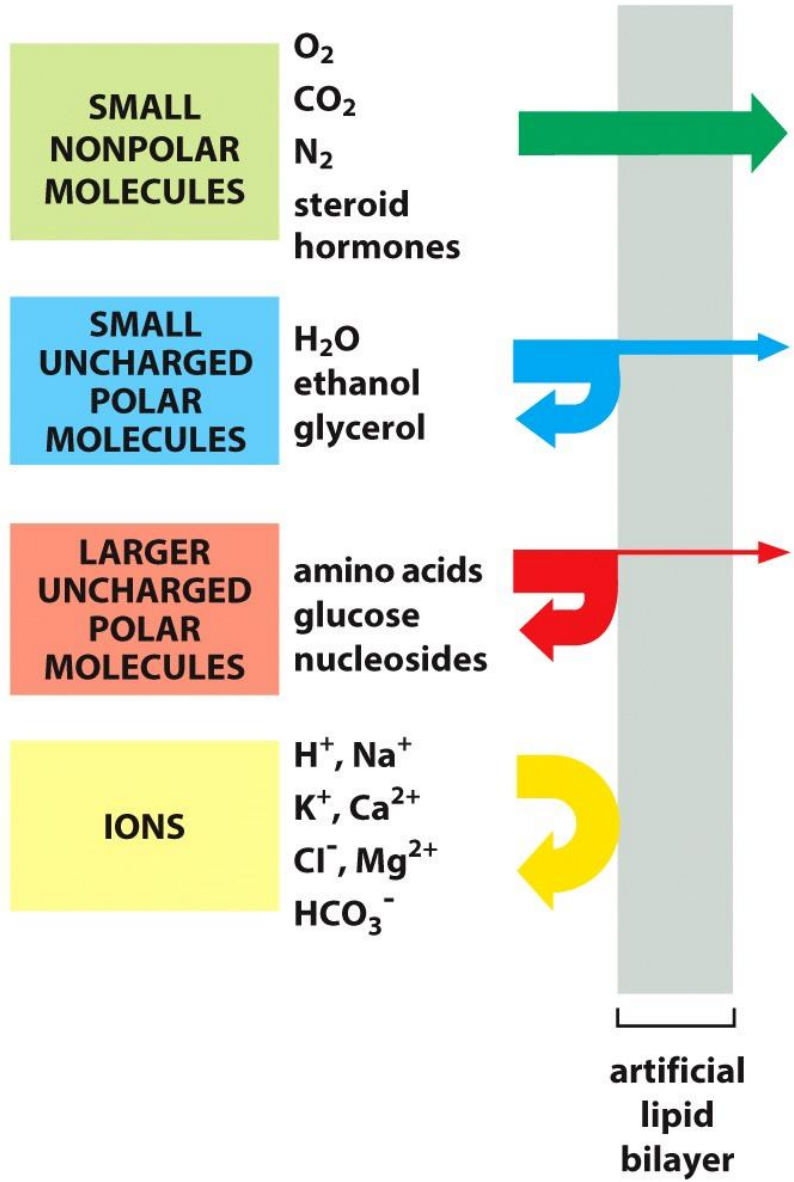
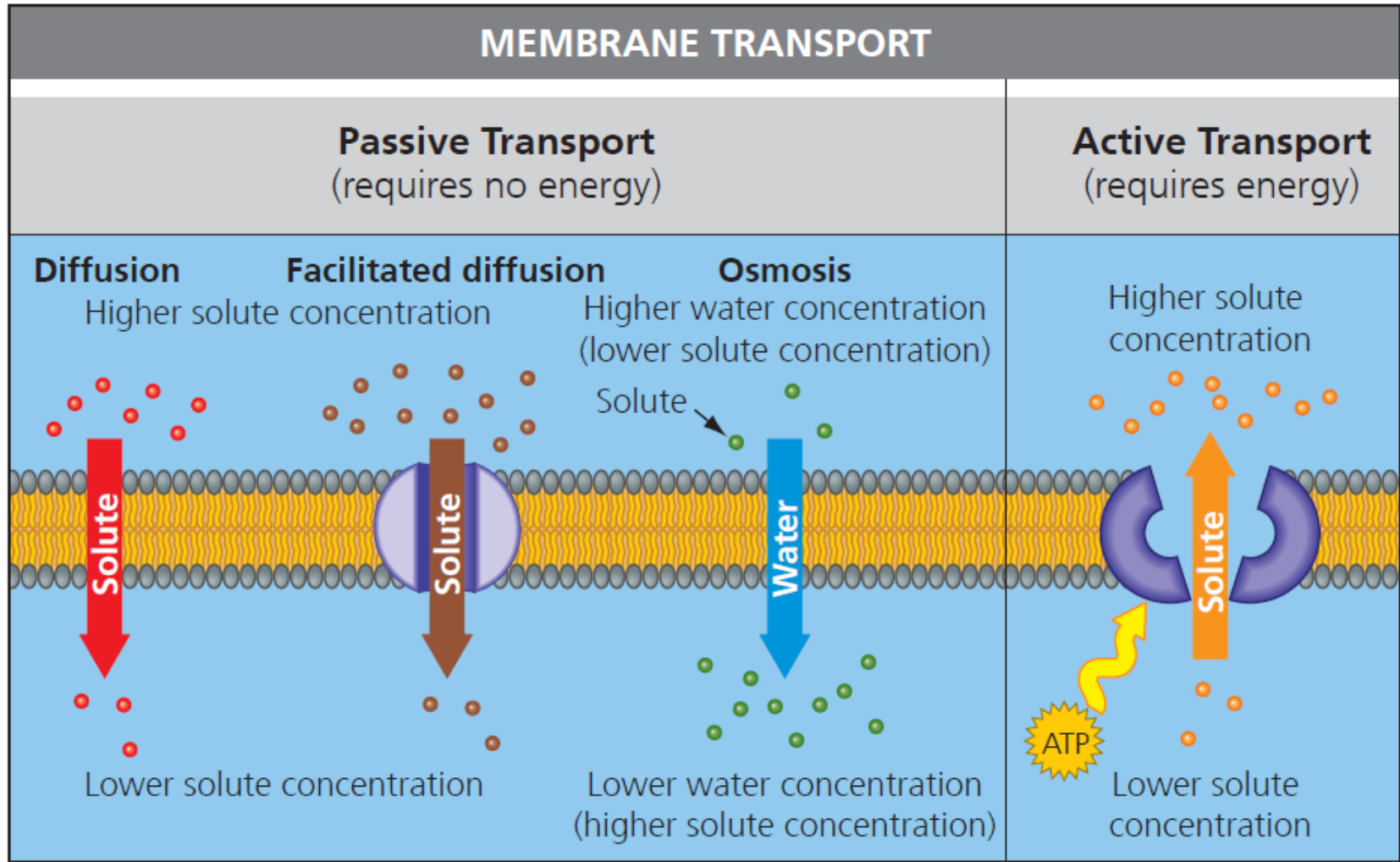
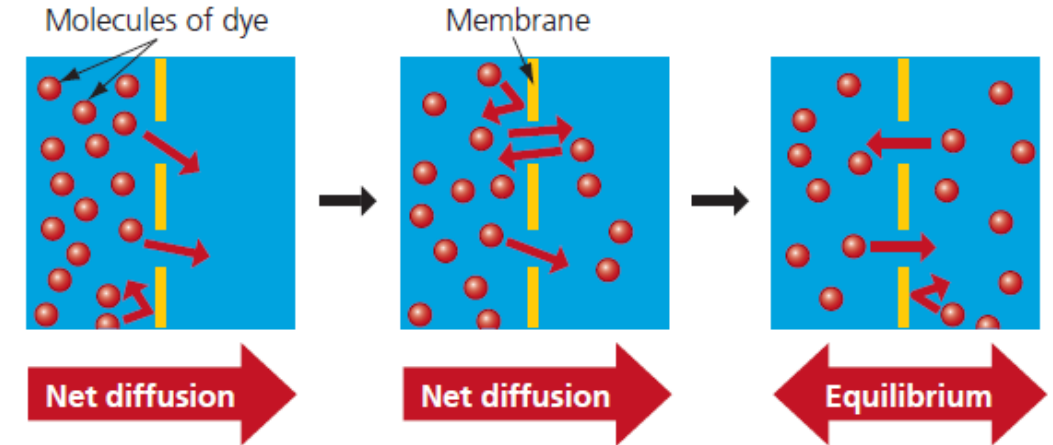


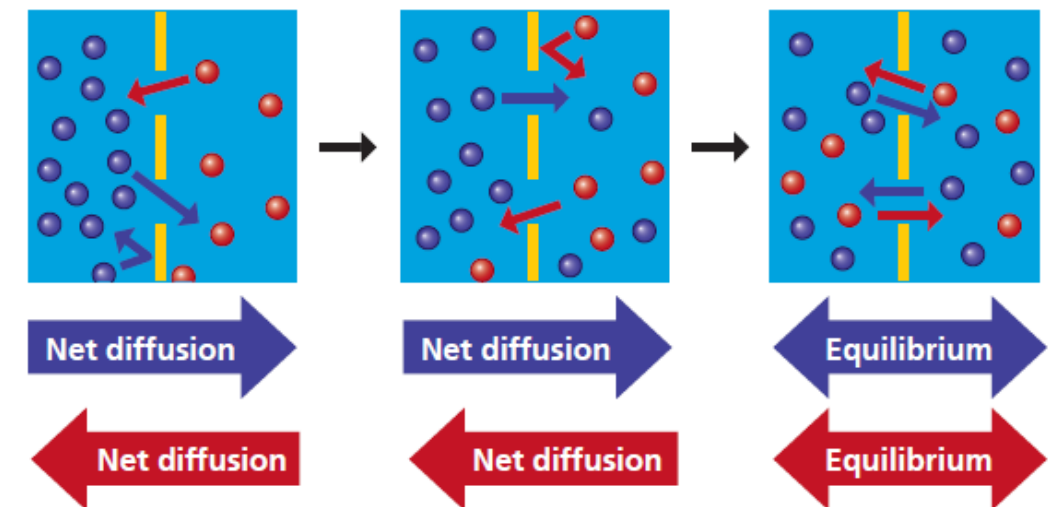
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Passive Transport: Diffusion across Membranes

- Molecules constantly vibrate and wander randomly.
- Diffusion is the movement of molecules spreading out evenly into the available space.
- “Passive” means “no input of energy”.
- Simple diffusion
- Facilitated diffusion



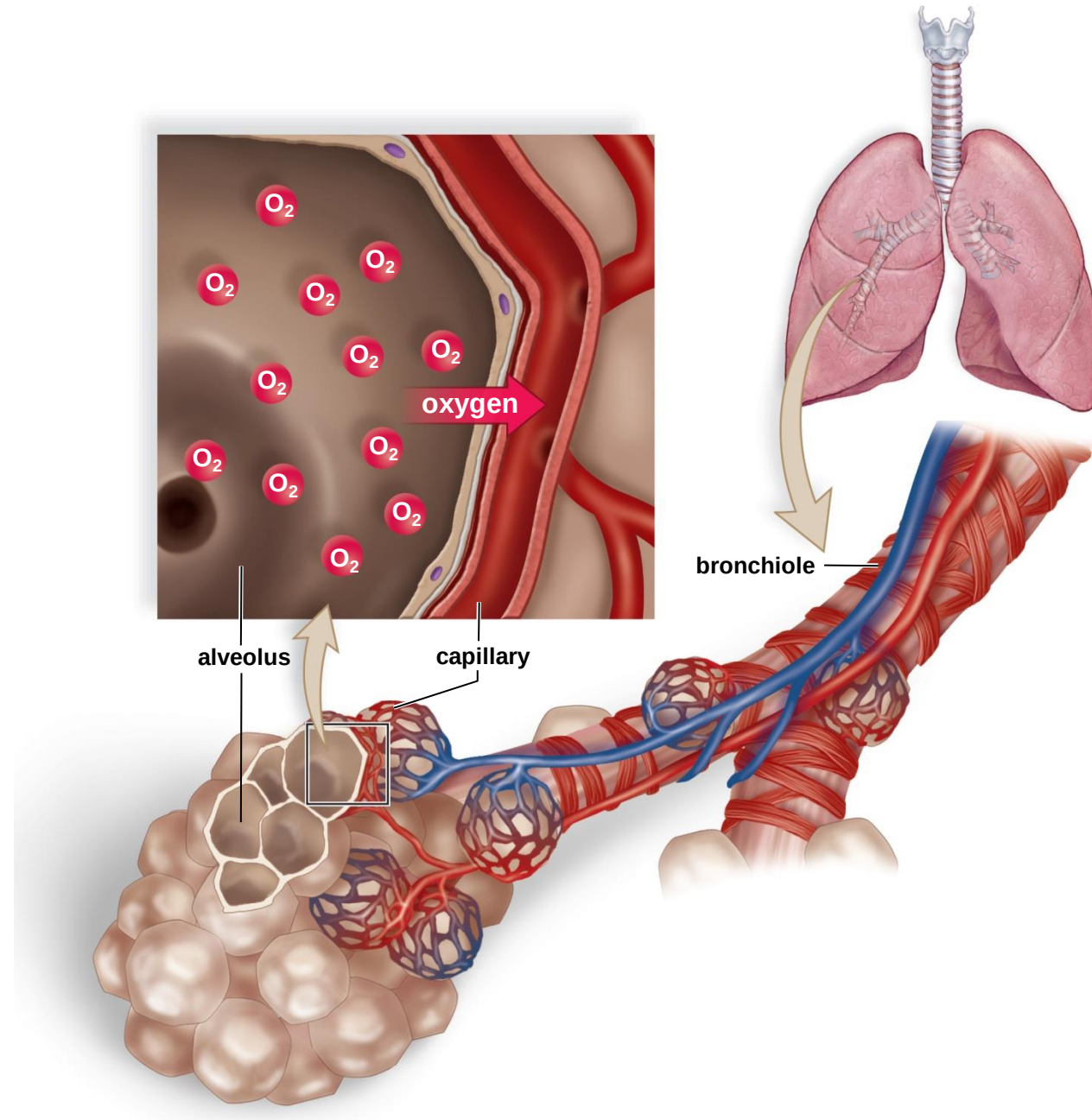
(a) **Passive transport of one type of molecule.** The membrane is permeable to these dye molecules, which diffuse down the concentration gradient. At equilibrium, the molecules are still restless, but the rate of transport is equal in both directions.



(b) **Passive transport of two types of molecules.** If solutions have two or more solutes, each will diffuse down its own concentration gradient.

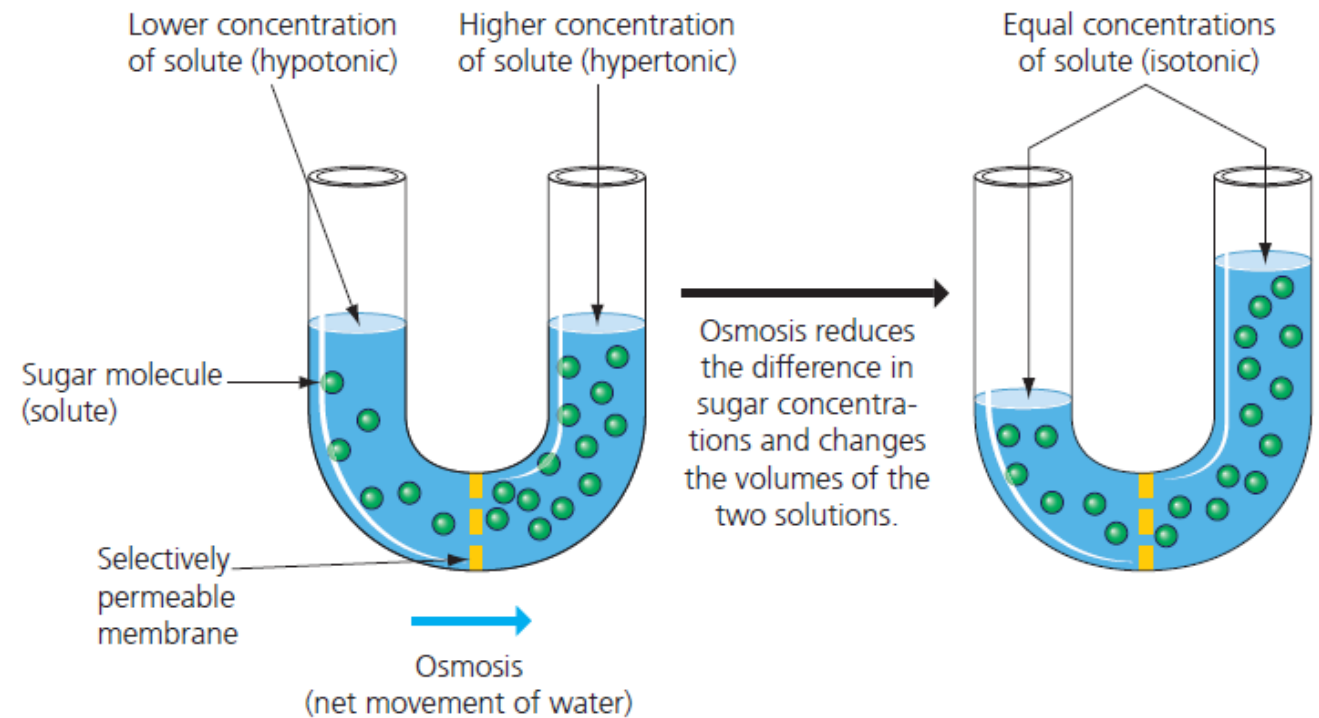
Simple and Facilitated Diffusion

- Simple diffusion
 - Gasses, lipid, water
- Facilitated diffusion, assisted transport by corridors
 - Substances that do not cross membranes spontaneously, or otherwise cross very slowly
 - Glucose, water, ions



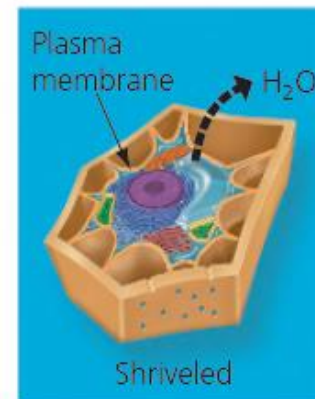
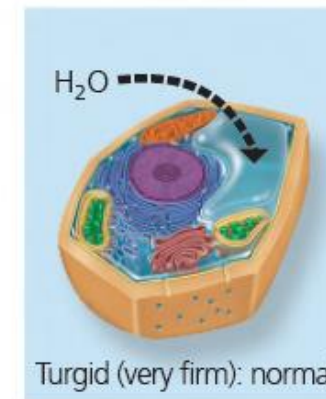
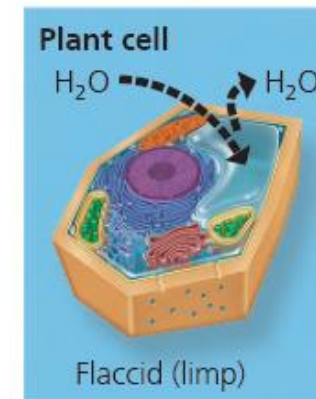
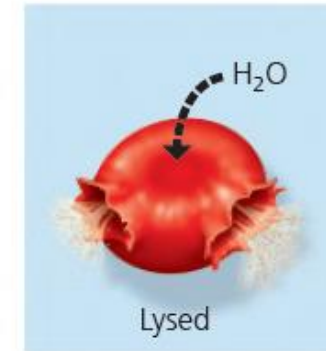
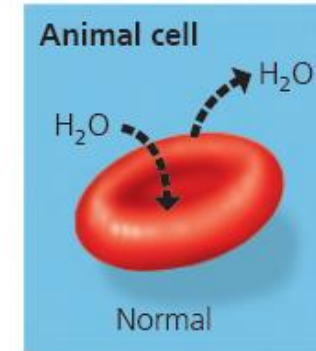
Osmosis (渗透作用)

- The diffusion of water across a selectively permeable membrane.
- Compared to another solution,
 - a **hypertonic** solution has a higher concentration of solute
 - a **hypotonic** solution has a lower concentration of solute
 - an **isotonic** solution has an equal concentration of solute



Water Balance in Animal and Plant Cells

- For an animal to survive a hypotonic or hypertonic environment, the animal must have a way to balance the uptake and loss of water, **osmoregulation** (渗透调节).
- **Plant** cells are healthiest in a **hypotonic environment** with a net inflow of water, which expands their cell walls (rigid) without bursting.



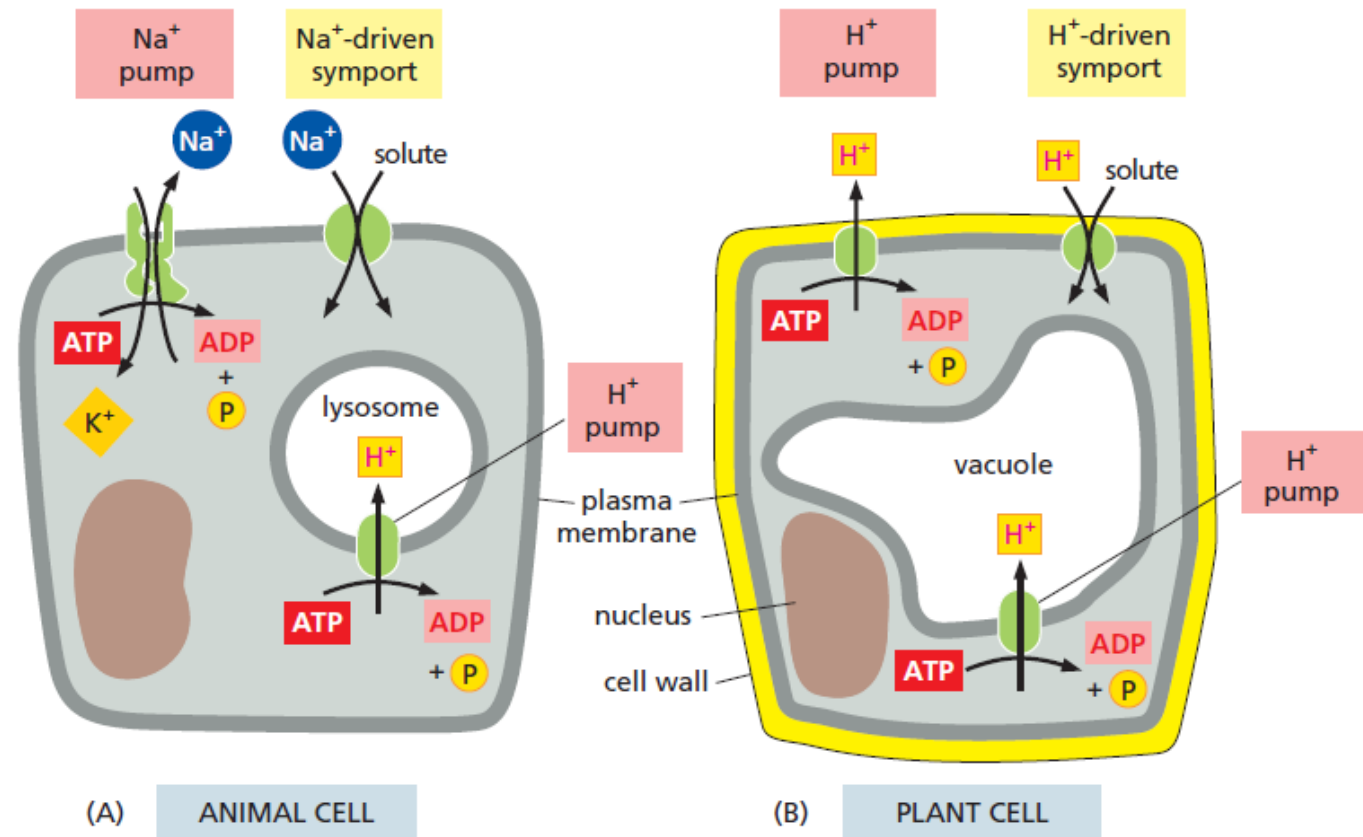
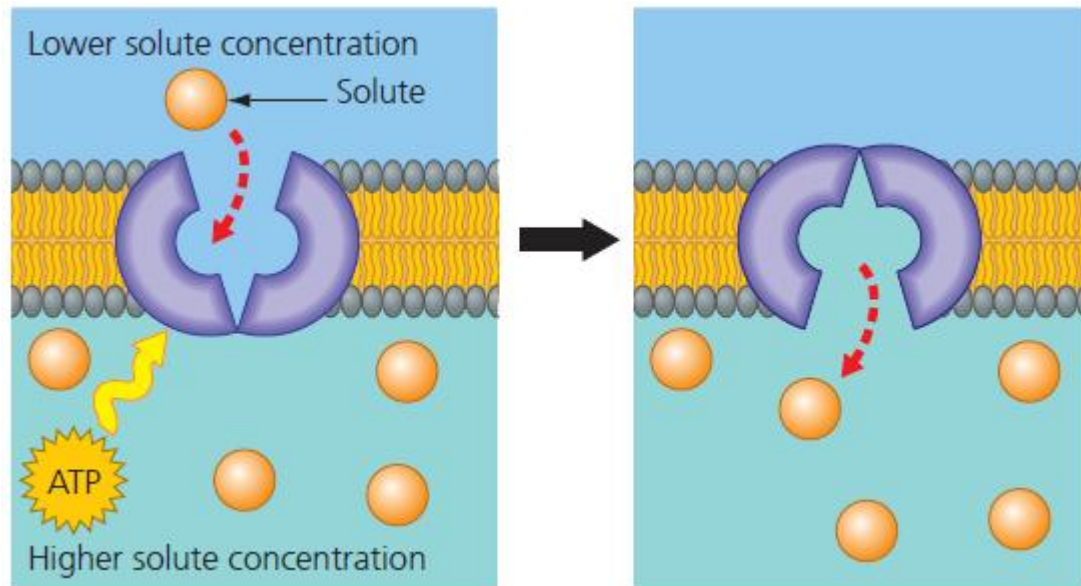
(a) Isotonic solution

(b) Hypotonic solution

(c) Hypertonic solution

Active Transport: The Pumping of Molecules across Membranes

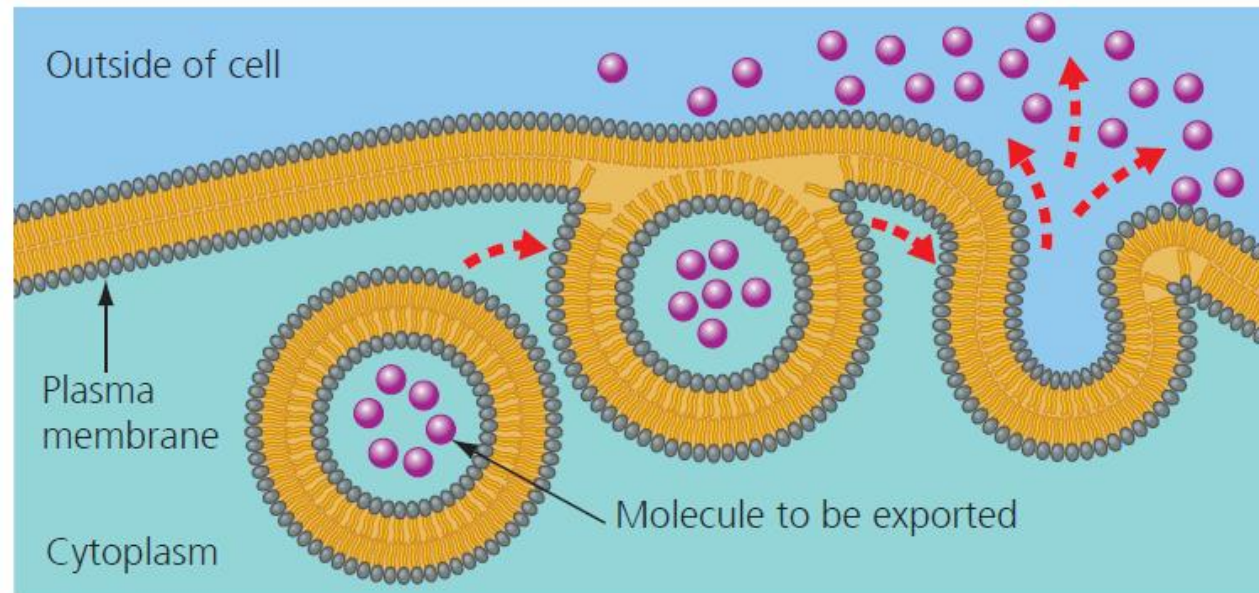
- “Active” means “expending energy”.
- Cellular energy is used to drive a transport protein that pumps a solute **against** the concentration gradient.



Exocytosis (胞吐) and Endocytosis (胞吞): Traffic of Large Molecules

- Exocytosis is the movement of materials **out of** the cytoplasm of a cell via membranous vesicles or vacuoles that fuse with the plasma membrane.
- In endocytosis, a cell takes material in via vesicles that bud inward.

▼ Figure 5.17 Exocytosis.



▼ Figure 5.18 Endocytosis.

